VICTORY 2027

ELECTRICITY AND MAGNETISM

LECTURE-06 PHYSICS

Shailendra sir



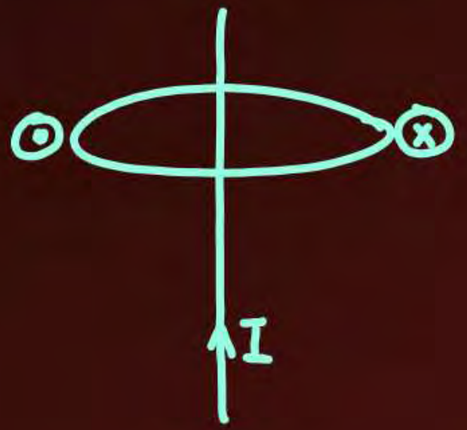
Topics To Be Covered

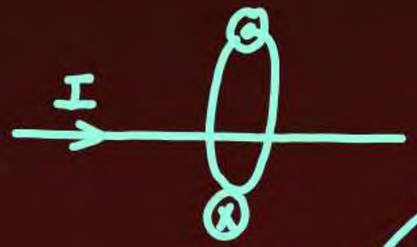
1. OERSTED'S EXPERIMENT
2. FIELD LINES DUE TO CURRENT IN A STRAIGHT WIRE
3. ELECTROMAGNET
4. QUESTIONS

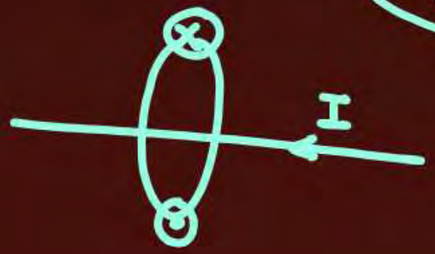

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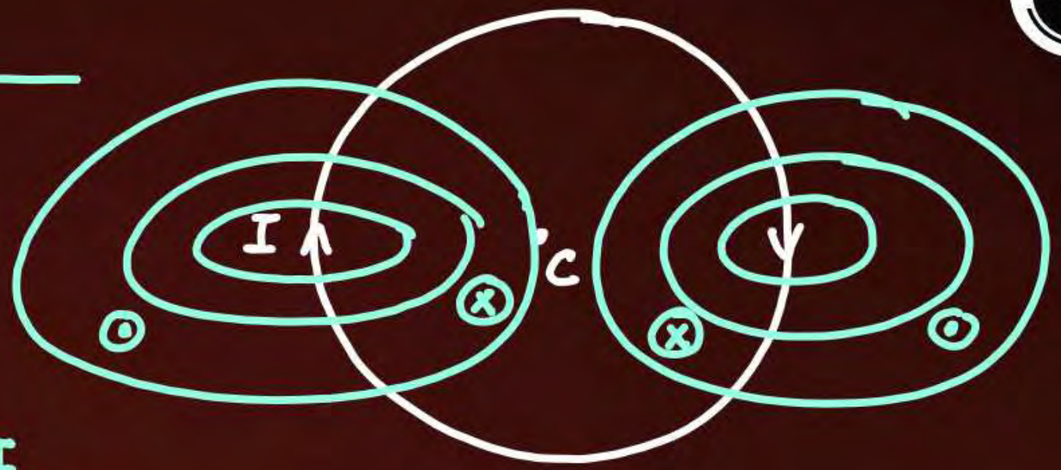


Revision



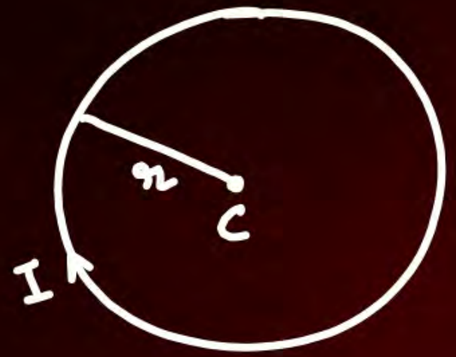




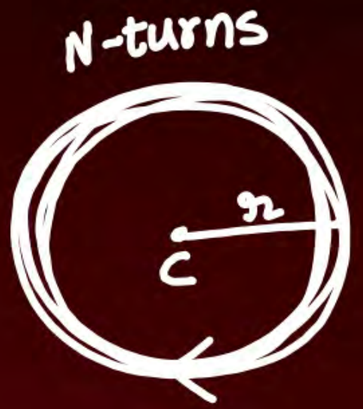


$$B_c \propto \frac{I}{r}$$

$$B_c = 2\pi \times 10^{-7} \frac{I}{r}$$



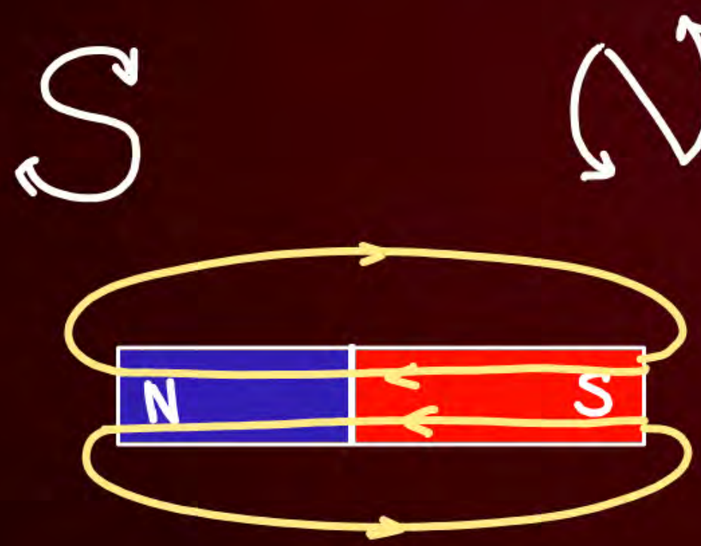
$$B_c = 2\pi \times 10^{-7} \frac{I}{r}$$

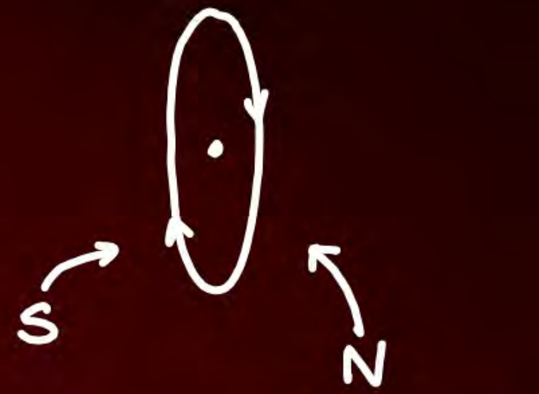


$$B_c = 2\pi \times 10^{-7} \frac{NI}{r}$$

Clock-Face Rule



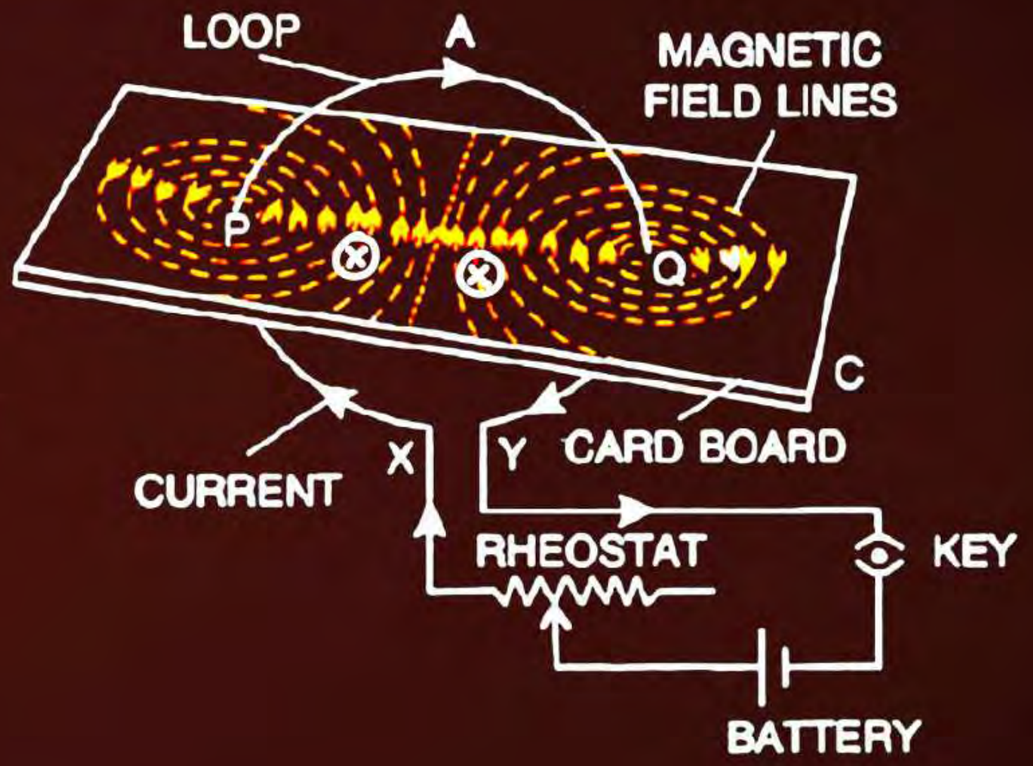




Magnetic Dipole

MAGNETIC FIELD DUE TO CURRENT IN A LOOP (OR CIRCULAR COIL)

- **Near the wire (P & Q)**: Field lines are almost circular.
- **Inside the loop:** Field lines are in the same direction.
- **Near the centre:** Field lines are nearly parallel → field is approximately uniform.
- **At the centre:** Field lines are along the axis of the loop and perpendicular to its plane.
- **Magnetic field strength increases when:**
  - **Current increases** $B \propto I$
  - **Number of turns increases** $B \propto N$
- The face from which field lines emerge → North pole.
- The face where field lines enter → South pole.
- Hence, the current-carrying loop behaves like a magnetic dipole.



SIMILARITIES BETWEEN A CURRENT CARRYING SOLENOID AND A BAR MAGNET





Similarities:

- A current-carrying solenoid produces a magnetic field similar to a bar magnet.
- When freely suspended, it aligns in the North–South direction.
- It shows magnetic attraction and attracts iron filings.

Dissimilarities:

- **Field strength:**
  - Solenoid → can be changed by changing current;
  - Bar magnet → cannot be easily changed.
- **Field direction:**
  - Solenoid → can be reversed by reversing current;
  - Bar magnet → cannot be reversed.

QUESTIONS

In Fig, A and B represent two straight wires carrying equal currents in a direction normal to the plane of paper inwards.



(a) Sketch separately the magnetic field lines produced by each current.

(b) What will be the magnetic field at mid point K of the line joining A and B. Give reason. $B_K = 0$

(c) How will the magnetic needle of compass rest if it is placed at the point K? N-S Direction

(d) What will be the effect on the magnetic field at the point K if the current in wire B is reversed? increase.



QUESTIONS

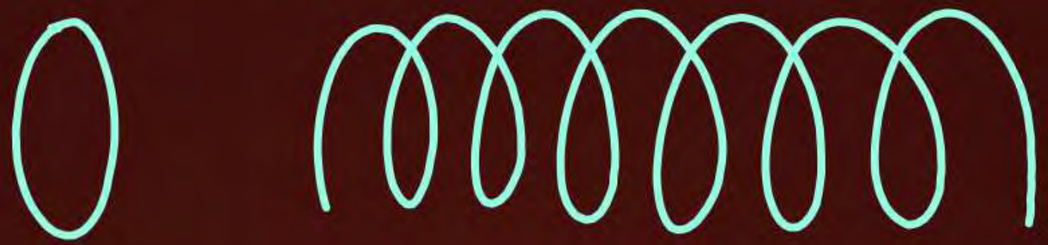
A coil wound around a piece of soft iron can become an electromagnet only when: [ICSE Sem-2, 2022]

Magnetism from Current

1. the circuit is open
2. a magnetic compass is present in the vicinity
3. a galvanometer is connected to the circuit
4. a current flows in the circuit.

QUESTIONS

When number of turns in the solenoid increases, magnetic field:



1. decreases
2. increases
3. may decrease or increase
4. cannot be determined.

$B \propto N$



QUESTIONS

When an electric current is passed through a conducting wire, a magnetic field is produced around it. This was proved by:

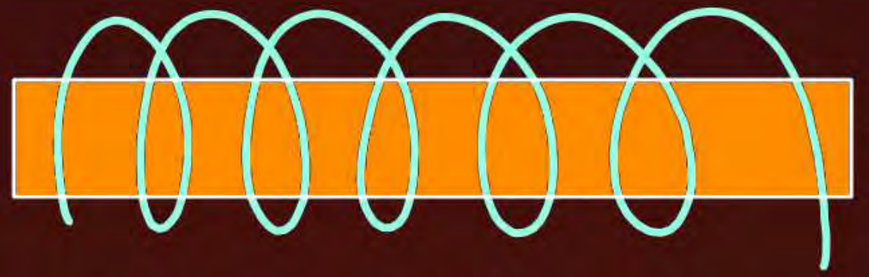
1. Oersted's experiment
2. Fleming's experiment
3. Newton's experiment
4. Lenz's experiment

QUESTIONS

When soft iron rod is placed in the core of solenoid, its magnetic strength:

1. increases
2. decreases
3. does not change
4. may decrease/increase

$n = \frac{N}{l}$



→ Soft Iron

Magnetism ↑↑

$B = 4\pi \times 10^{-7} n I = \mu_0 n I$

$B' = \mu_r (4\pi \times 10^{-7} n I) = \mu_r \mu_0 n I$



QUESTIONS

Electromagnet can be demagnetised by:

1. passing the current
2. switching off the current
3. decreasing the current
4. increasing the number of turns.

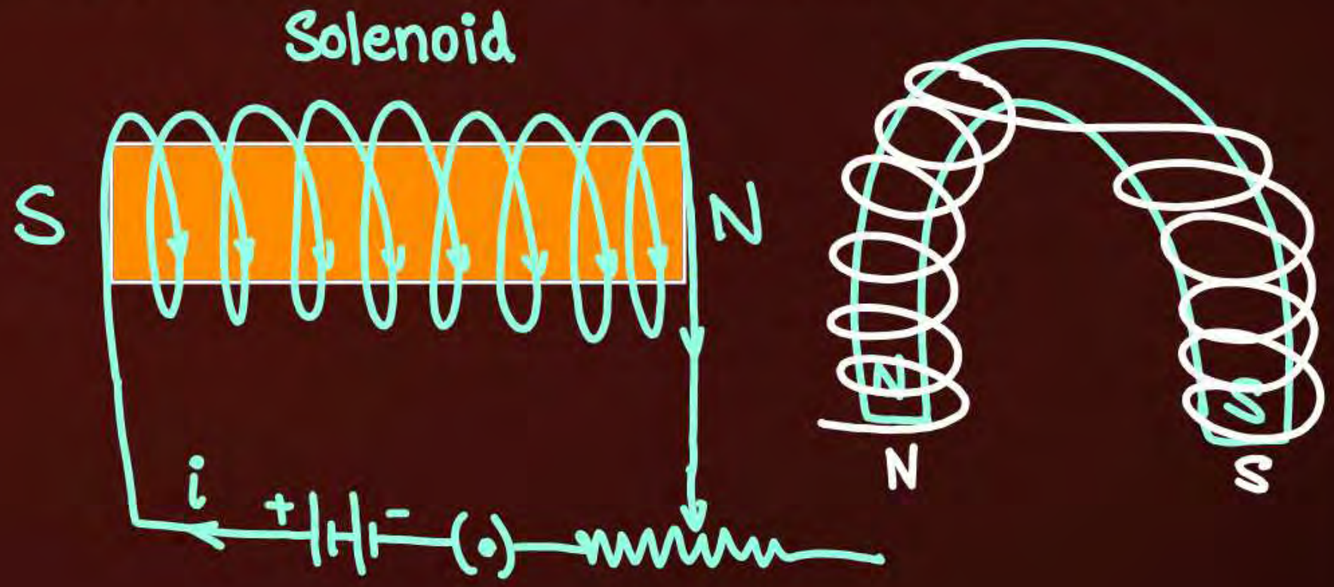
ELECTROMAGNET

An electromagnet is a temporary strong magnet made by passing current in a coil wound around a piece of soft iron. It is an artificial magnet.

An electromagnet can be made in any shape, but usually the following two shapes of electromagnet are in use:

(1) **I-shape (or bar) magnet**

(2) **U-shape (or horse-shoe) magnet**





SOFT IRON v/s HARD IRON

Soft Iron

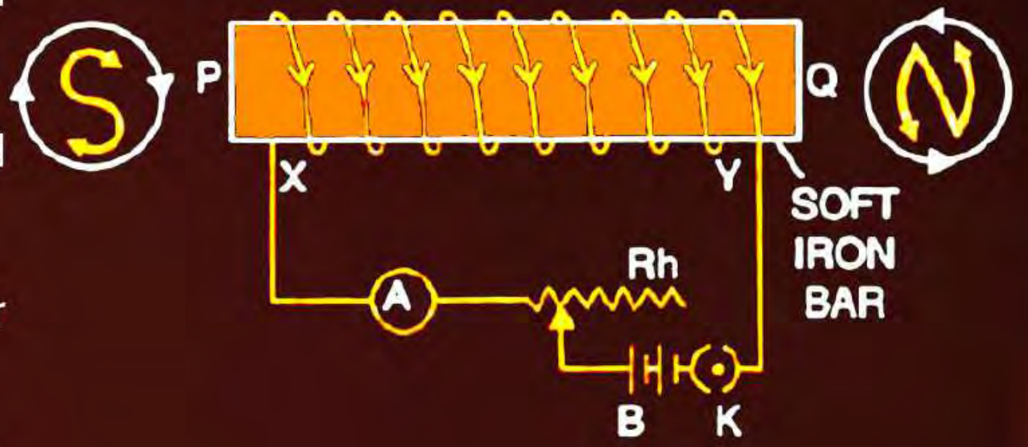
- **Magnetic Properties**: Easily magnetized and demagnetized.
- **Hysteresis Loop**: Narrow hysteresis loop.
- **Remanence**: Low ability to retain magnetism.
- **Applications**: Used in electromagnets, which need to be switched on and off.
- **Composition**: Typically contains very little carbon.

Hard Iron (or Steel)

- **Magnetic Properties**: Difficult to magnetize and demagnetize, but retains magnetism once magnetized.
- **Hysteresis Loop**: Broad hysteresis loop.
- **Remanence**: High ability to retain magnetism.
- **Applications**: Used for making permanent magnets.
- **Composition**: Hard iron often refers to materials like steel, which contain carbon and other elements that increase its hardness and ability to become a permanent magnet.

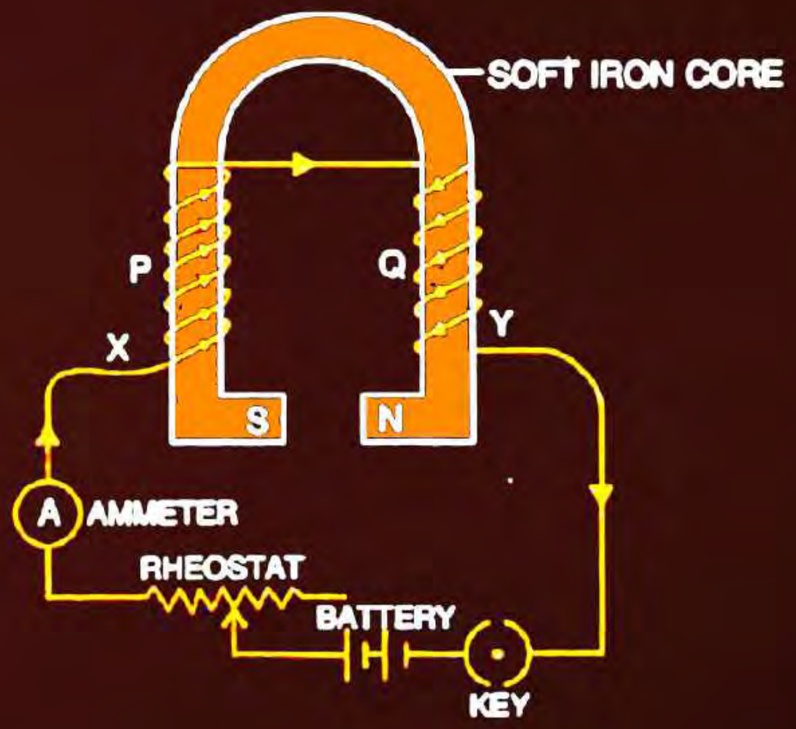
Construction of I-shaped (or bar) electromagnet

- A soft iron bar PQ is used as the core, with an insulated copper wire wound around it as a solenoid.
- The coil is connected to a battery, ammeter, rheostat and key in series.
- When the key is closed, current flows and the iron bar becomes a magnet.
- P → South Pole (S) because current is clockwise at P.
  Q → North Pole (N) because current is anticlockwise at Q.
- The bar remains magnetic only while current flows. When current is switched off, it loses its magnetism because soft iron has low retentivity.
- Therefore, it is a **temporary magnet** (electromagnet).
- **Use**: I-shaped electromagnets are commonly used in relays.



Construction of U-shaped (or horseshoe) electromagnet

- A U-shaped soft iron bar is used as the core.
- An insulated copper wire is wound around both arms in the form of coils.
- The winding directions on the two arms are opposite when viewed from their ends.
- The coil is connected to a battery, ammeter, rheostat and key in series.
- When the key is closed, current flows through the coil and the iron core becomes a horseshoe electromagnet.
-
- The two arms develop opposite magnetic poles (N and S).



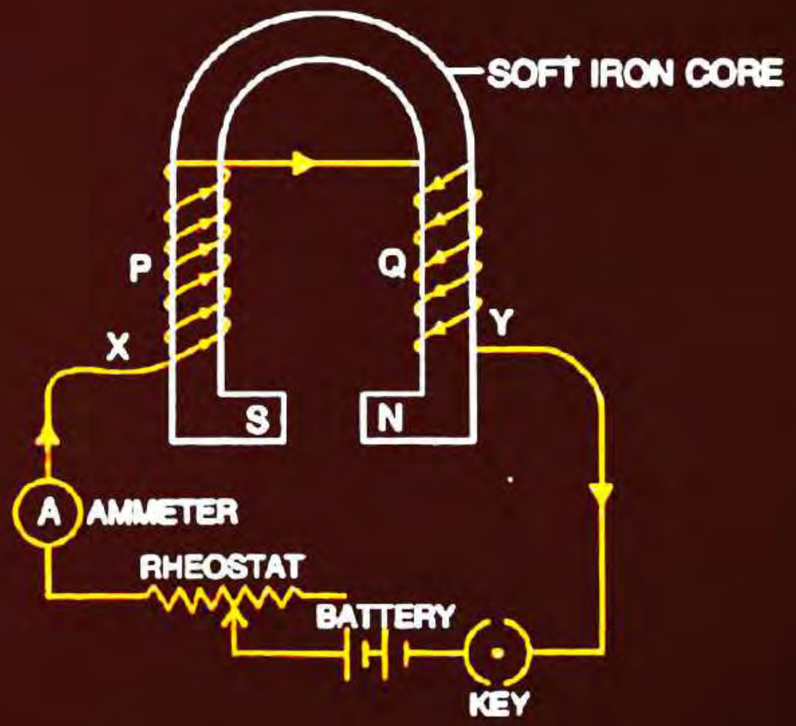
Construction of U-shaped (or horseshoe) electromagnet

- When the key is closed, current flows through the coil.

- P → South Pole (S) because current is clockwise.
- Q → North Pole (N) because current is anticlockwise.

- Opposite poles produce a strong magnetic field in the gap.
-
- When current is switched off, the magnetic field disappears as soft iron loses its magnetism. Hence, it is a **temporary magnet.**

- **Uses:**
  - ✓ D.C. motors
  - ✓ A.C. generators
  - ✓ Electric bells



ELECTROMAGNET



Note:

- For sending current in the coil, the source must be the d.c. source (i.e., battery).
- With an **a.c. source of frequency 50 Hz** although the soft iron core gets magnetised as long the current is passed, but its polarity at the ends changes 50 times in each second.

Ways of increasing the magnetic field of an electromagnet

$B \propto N$

(1) by increasing the number of turns of winding in the solenoid, and
(2) by increasing the current through the solenoid.

$B \propto I$

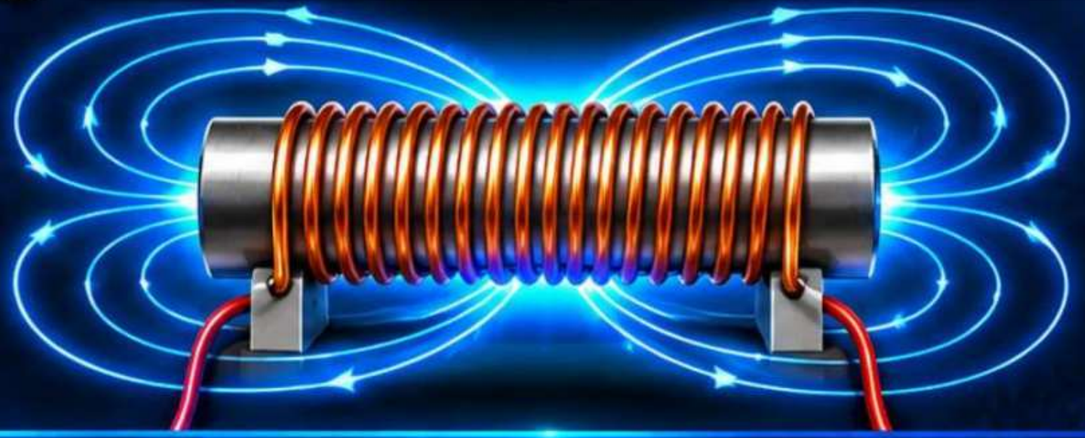
PERMANENT MAGNET

- A permanent magnet retains its magnetism for a long time.
- It is made by magnetising steel using a coil carrying electric current.
- Steel has high retentivity, so it does not lose its magnetism easily.
- Hence, unlike soft iron, steel becomes a permanent magnet after magnetisation.
- **Uses:**
  - Galvanometer → to check presence of current
  - Ammeter → to measure current
  - Voltmeter → to measure voltage.
  - Magnetic compass → to detect Magnetic field.

COMPARISON OF AN ELECTROMAGNET WITH A PERMANENT MAGNET

Electromagnet

1. Made of **soft iron**.
2. Produces a **magnetic field only when current flows** in its coil (temporary magnet).
3. Magnetic field strength **can be changed**.
4. Magnetic field of an electromagnet **can be very strong**.
5. The polarity of an electromagnet **can be reversed**.
6. Can be **easily demagnetised** by switching off the current.

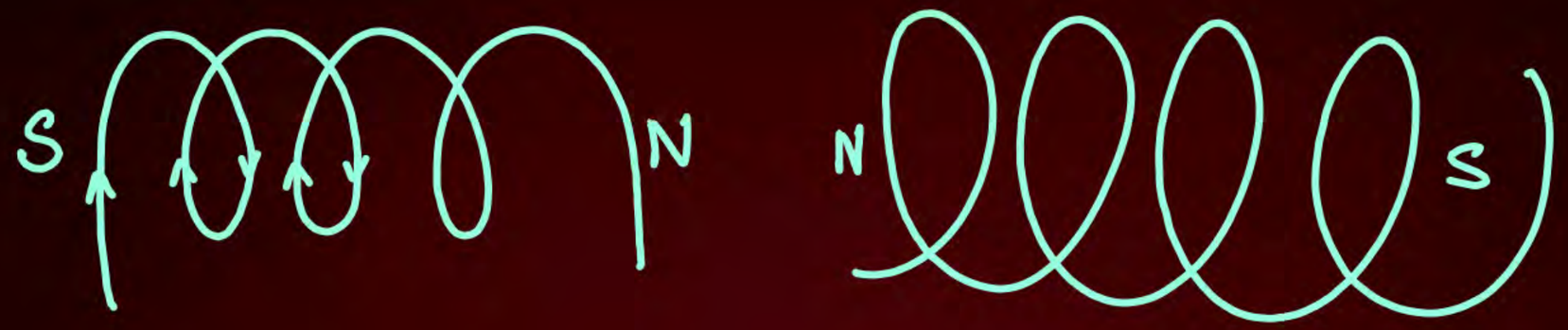


VS

Permanent Magnet

1. Made of **steel**.
2. Produces a **permanent magnetic field**.
3. Magnetic field strength **cannot be changed**.
4. Magnetic field of a permanent magnet **is not very strong**.
5. The polarity of a permanent magnet **cannot be reversed**.
6. **Cannot be easily demagnetised**.





Rheostat → Variable Resistance.



USES OF ELECTROMAGNET

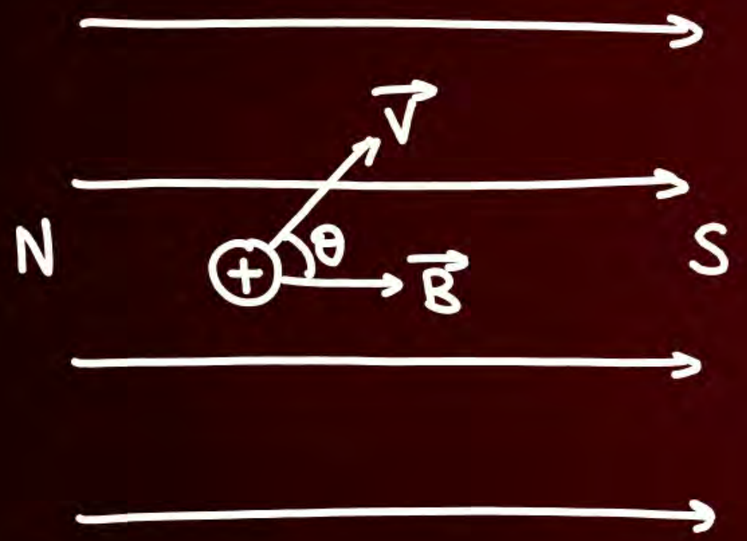
- Lifting heavy iron objects like scrap, girders and plates; switching off current releases them.
- Loading furnaces with iron.
- Separating iron from ores and other materials.
- Removing iron pieces from wounds.
- Scientific research to study magnetic properties of substances.

- Used in electrical devices such as:
  - ✓ Electric bells
  - ✓ Telegraph
  - ✓ Electric trams
  - ✓ Electric motors
  - ✓ Relays
  - ✓ Microphones
  - ✓ Loudspeakers

Concept of Force in Magnetism

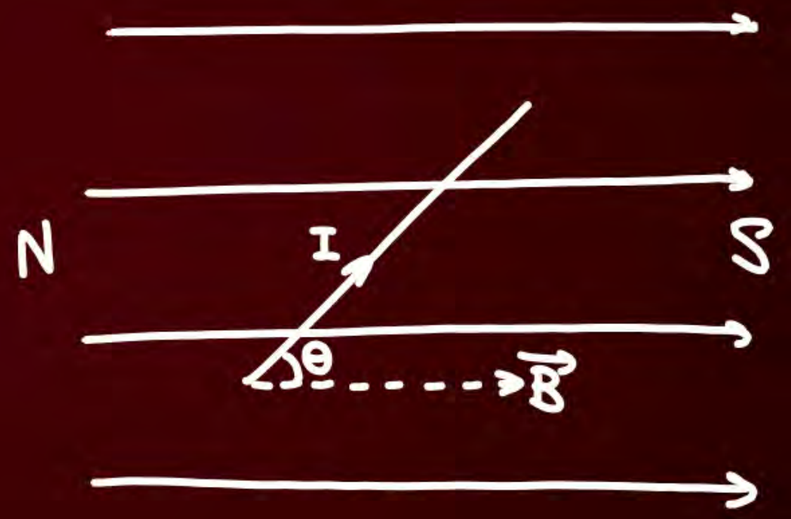
① When charge moves in MF

$$F = qvB\sin\theta$$



② When current flows in conductor in MF

$$F = I\ell B\sin\theta$$



$\sin 0^\circ = 0$

$\sin 30^\circ = \frac{1}{2}$

$\sin 45^\circ = \frac{1}{\sqrt{2}}$

$\sin 60^\circ = \frac{\sqrt{3}}{2}$

$\sin 90^\circ = 1$

$\sin 180^\circ = 0$

Direction of Force

↳ Flemming's Left Hand Rule.

Fore Finger → $\vec{B}$

Middle " → I

Thumb → F

SSS

Father → Thumb → force

Mother → Fore finger → MF

Child → Middle finger → I

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

Thank You